

85% charge efficiency. The tin deposits on the iron electrode suppress iron evolution.

The use of ionic liquids like 1-ethyl-3-methylimidazolium L-(+)-lactate (EML) as an electrolyte additive plays a major role in suppressing hydrogen release and anode passivation. EML has many desirable properties, such as high ionic conductivity, low vapour pressure, high chemical and thermal stability, and it is non-flammable. Their structure can be varied by incorporating different cations and anions.

The efficacy of EML as a hydrogen evolution inhibitor is most optimal when 1.0mM EML is added to the alkaline electrolyte of 6.0 M KOH solution. At this optimal concentration, it has been found that the hydrogen evolution rate is drastically reduced and is 35 times less than if no EML is used. Charge transfer resistance at the electrode during the hydrogen evolution reaction remarkably increases (about 50 times) with the addition of EML, due to adsorption of EML molecules on the cathodic sites of the iron electrode forming passivation layers of iron lactate. This passivation layer leads to a decrease in the cathodic reaction, resulting in minimal hydrogen evolution.

The surface morphology of the iron electrode, when examined by SEM and EDX, establishes that the iron electrode in blank electrolyte, i.e., 6.0 M KOH, is composed of a compactly stacked passive layer only, whereas if 1.0 M EML is used, the same surface becomes smooth without the formation of any passive layer. The addition of EML enhances the iron-air battery capacity more than three times as can be seen from Fig. 13.

As can be seen from Fig. 14, average capacity retention is about 58% after 385 cycles if no EML is used. But with the addition of 1.0 M of EML, capacity retention increases to about 94% even after 1000 cycles.

Lithium batteries are very good at cycling many thousands of times. However, if there is a battery that discharges for a week and then charges for a week, the theoretical maximum number of cycles one could get out of the battery is 26 per year. So, if the battery could provide 500 cycles, it would have a 20-year life. If the priority is a low-cost battery, there is no need to consider thousands of cycles like a Li-ion battery. The key is to make the right trade-offs in pursuit of the attributes that are important.

At present, grids are provided with storage batteries that allow them to smooth out multiple days of intermittency associated with renewable power generation driven by weather. However, these bat-



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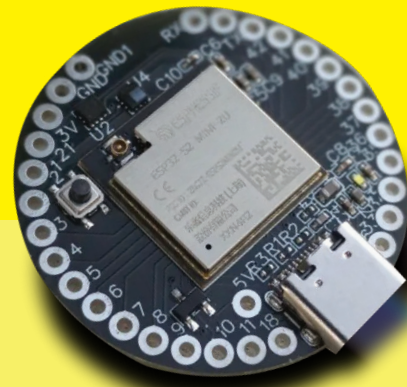
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